

Your Benchmark Is Not Saturated: Reviving Multiple-Choice Evaluation with Answer Pooling

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INTRODUCTION

MCQs benchmarks (e.g., GPQA, MMLU-Pro,...) are the most commonly used way to evaluate new LLMs capabilities.

Q: Capital of France?

A. Berlin C. Paris
B. Madrid D. Rome

But, they are running out of room....!

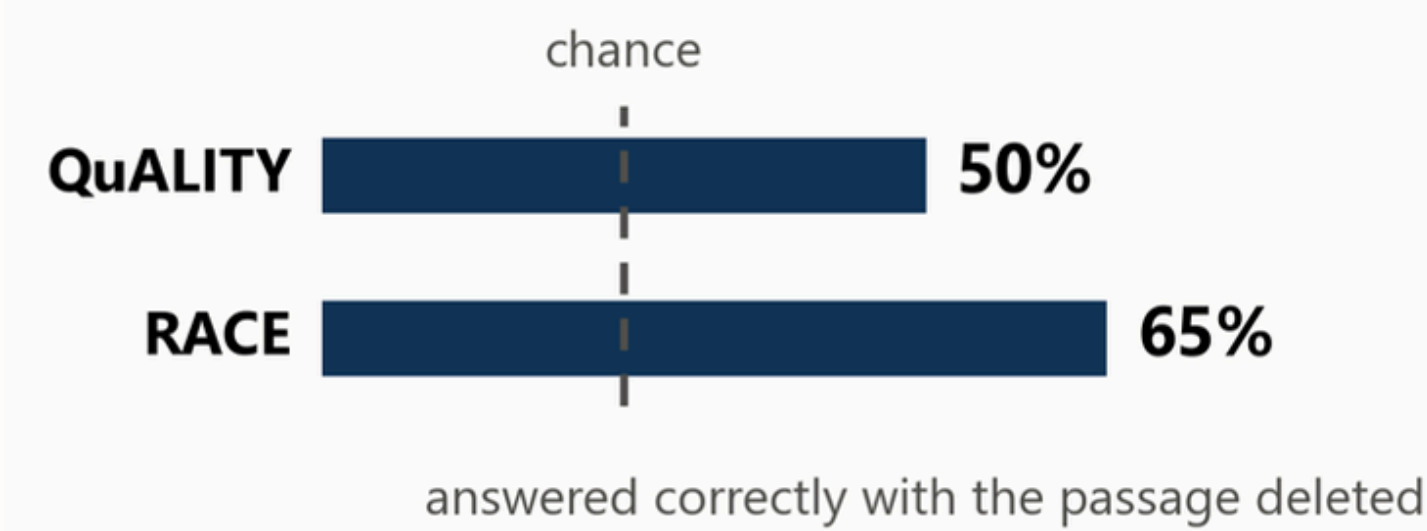
PROBLEM STATEMENT

1 They saturate FAST!

Building a new benchmark takes months.
Models close the gap in weeks.

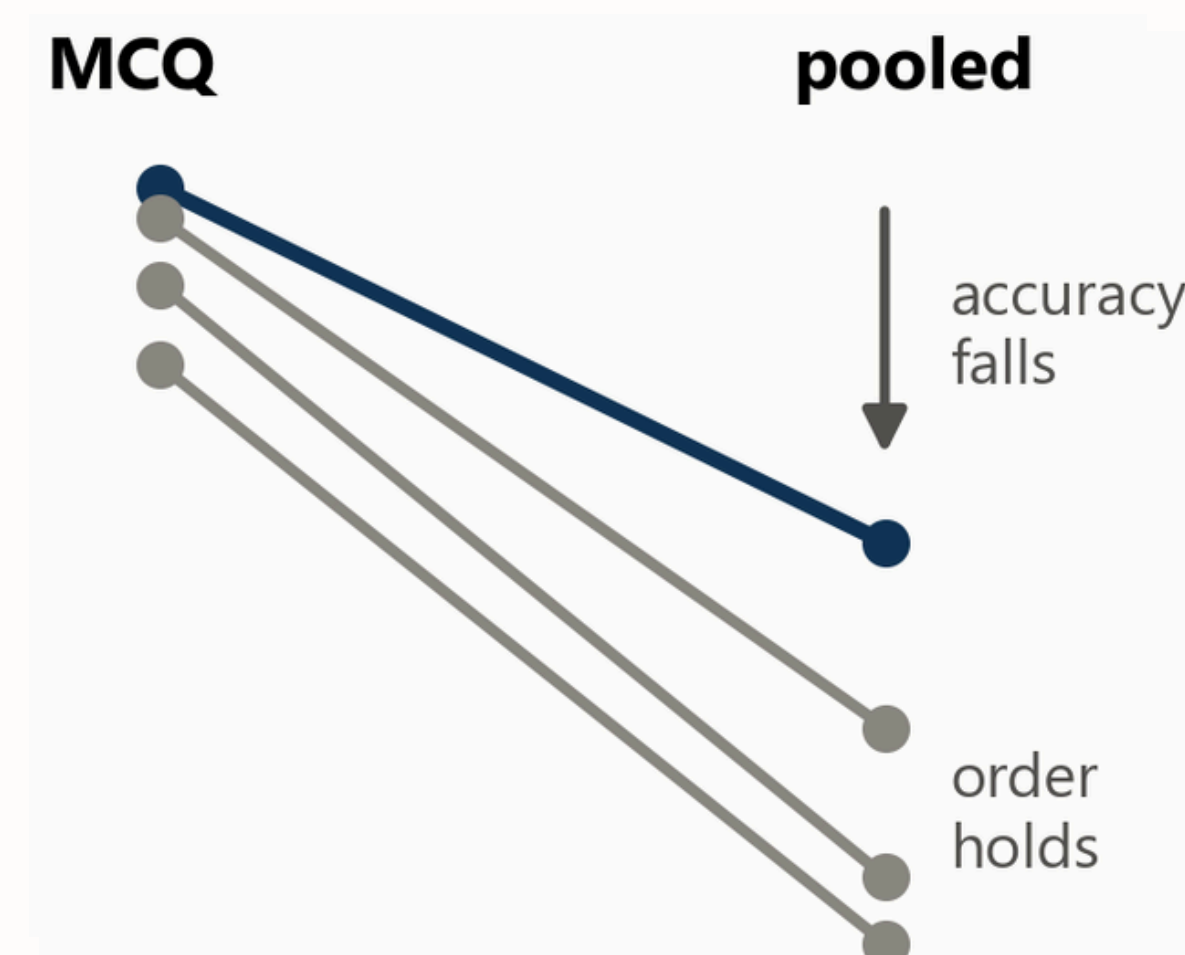


2 >50% of the questions never needed the input to answer correctly!



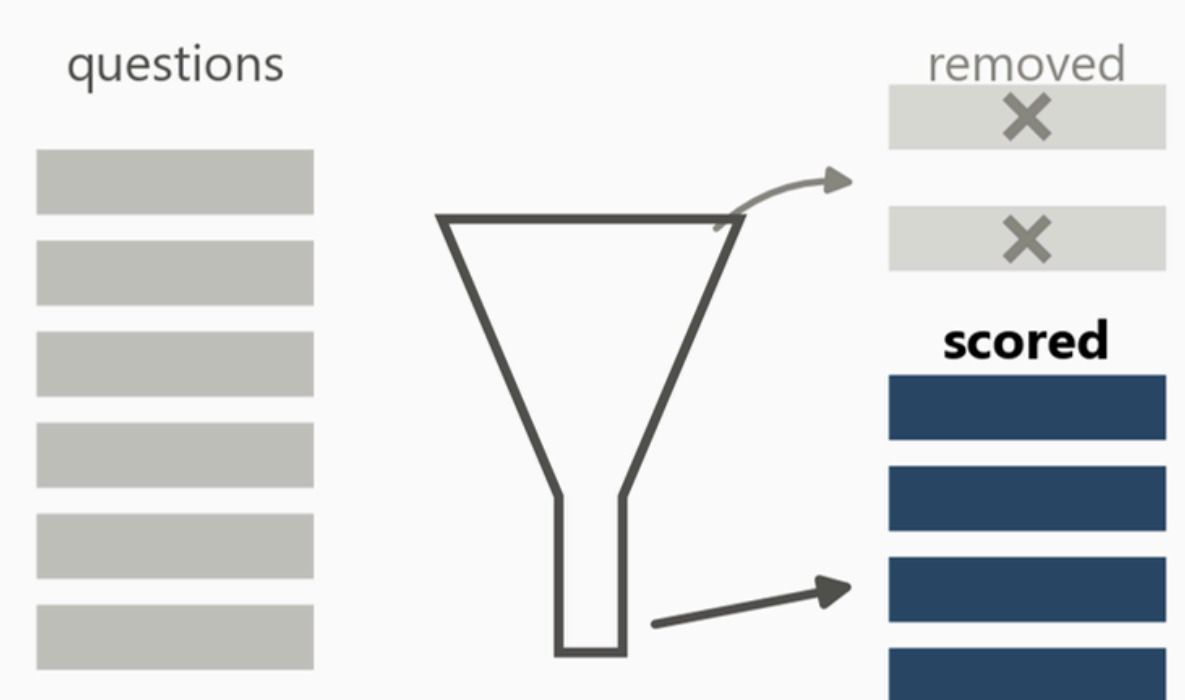
CONTRIBUTIONS

1 Benchmark de-saturation transform: a de-saturation transform for any MCQ benchmark, with zero human work



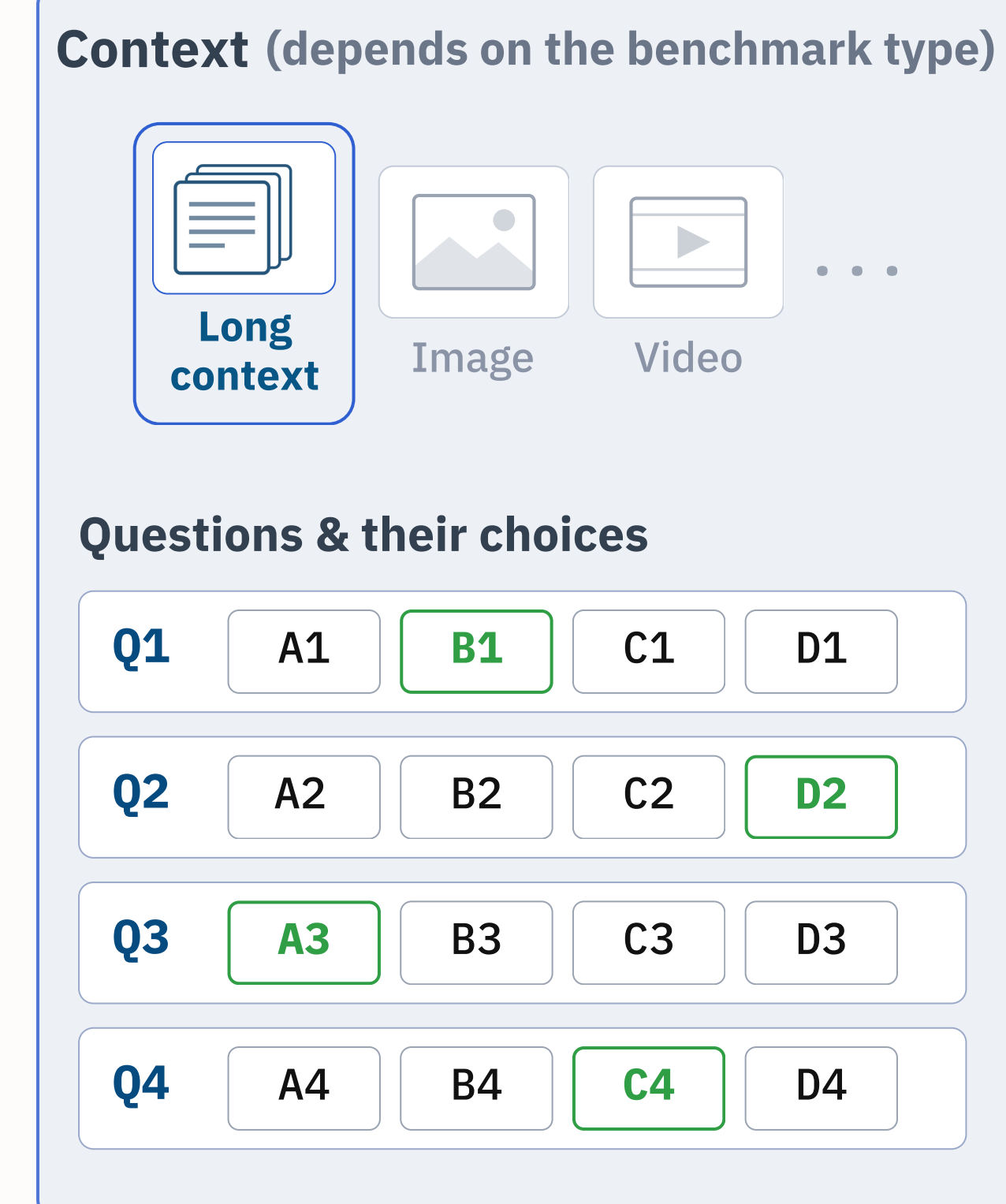
2 A filtering mechanism for MCQ benchmarks:

- 1) solvable without the context
- 2) ambiguous once pooled
- 3) separable by style



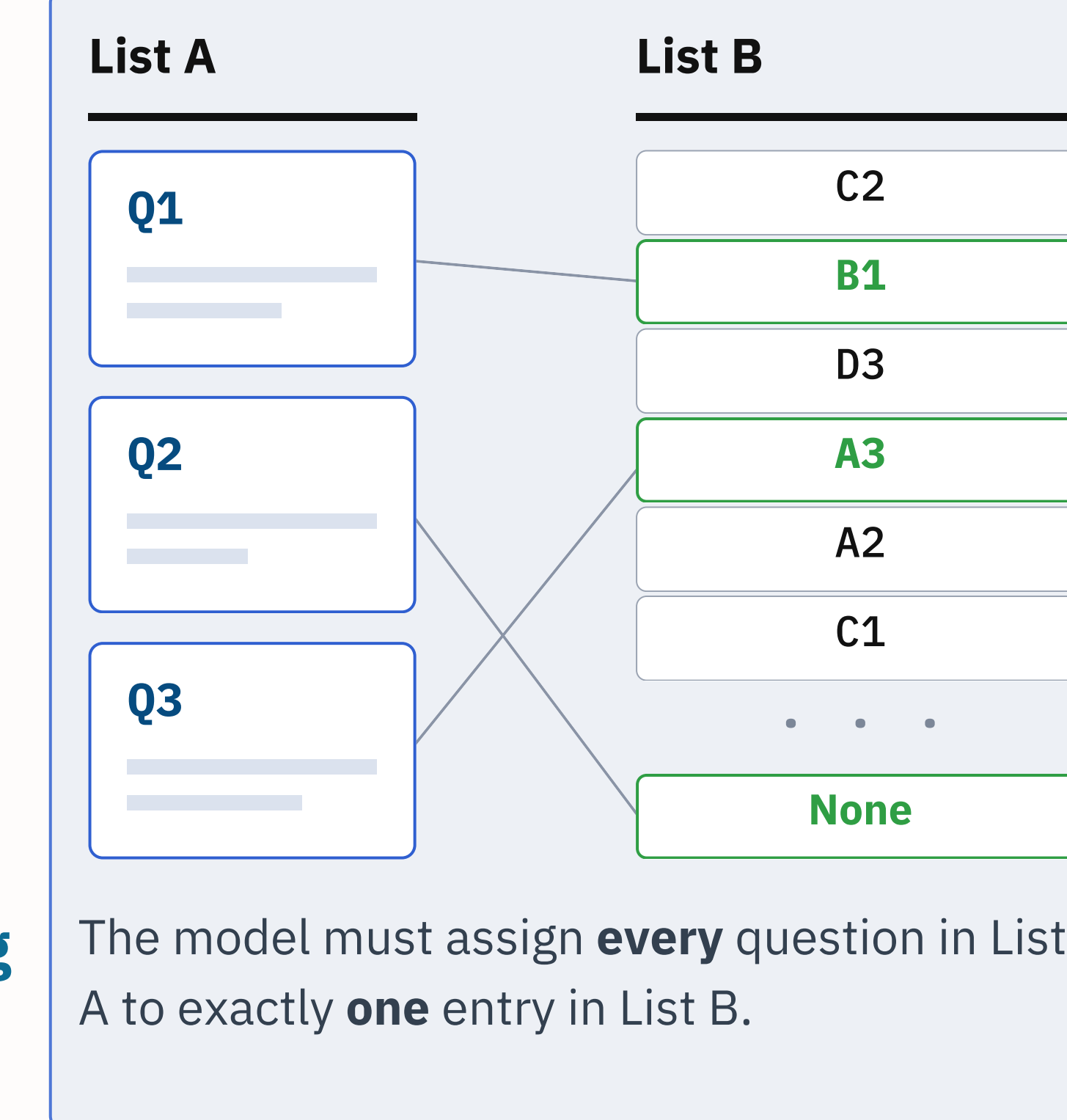
METHODOLOGY

Input (MCQs Benchmark)



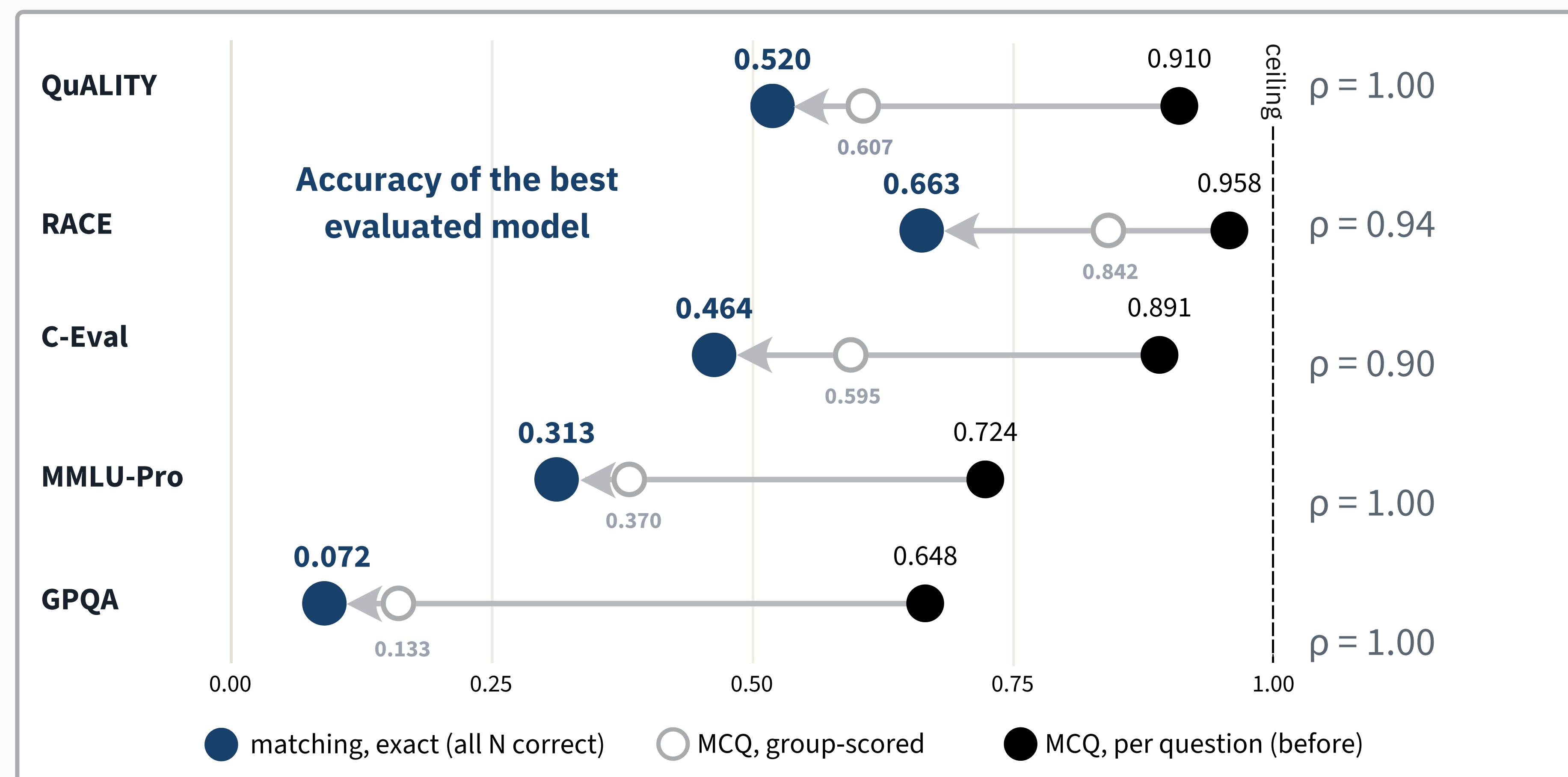
1. **Filtering**
Drop questions solvable without the context
Q4 dropped
2. **Withholding**
With a fixed ratio p , replace the correct choices with None
Q2 → None
3. **Pooling + Shuffling**
All choices of the kept questions

Our Formulation (Matching Task)



The transform. Filter out context-free questions, withhold a fraction p of golds, pool the rest into one shared list, and ask for the full assignment.

RESULTS (BENCHMARKS DESATURATION)

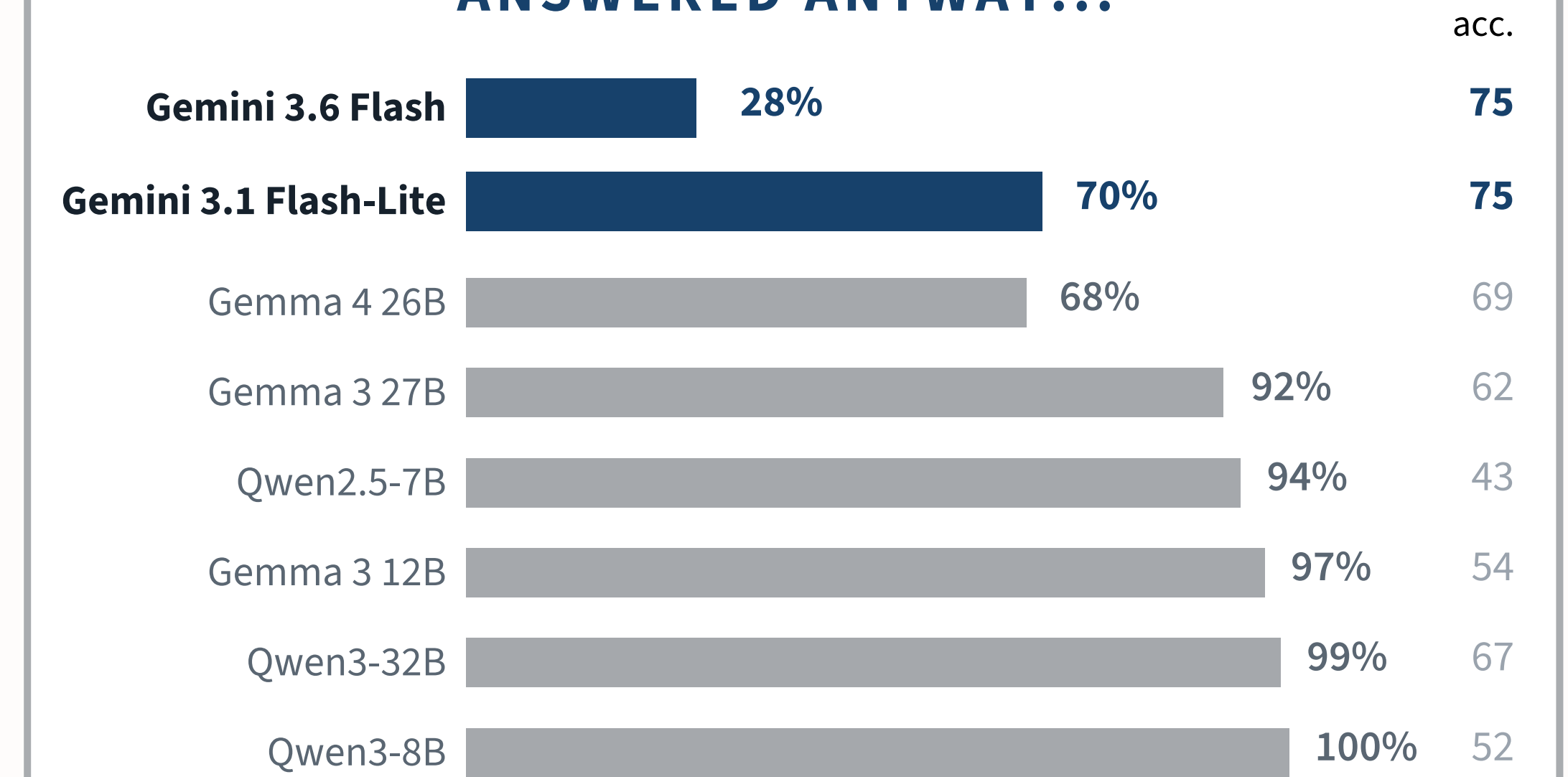


Accuracy of the best evaluated model before and after deasturation.

Group scoring removes part of the ceiling, matching removes the rest, and ρ (Spearman correlation with the original MCQ ranking) stays high across the benchmarks, proving the transform preserves the models rank.

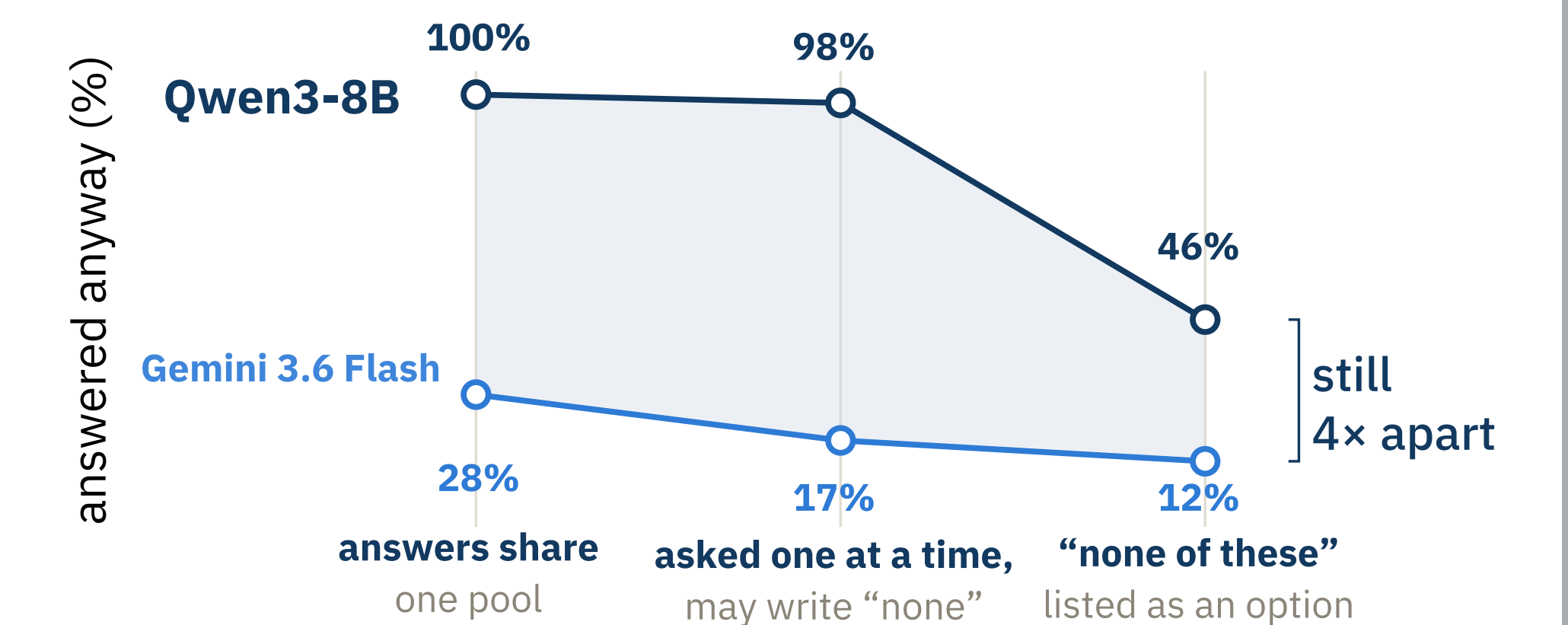
RESULTS (ABSTENTION SCORES)

QUESTIONS WITH NO CORRECT ANSWER, ANSWERED ANYWAY...



The two highlighted models are tied on accuracy yet 2.5× apart here!
Qwen3-8B declined only 2 of 900!

THE SAME MISSING QUESTIONS, ASKED THREE WAYS



The same questions, asked three ways, give three different numbers.

WHAT TO DO WITH THIS?

- ➔ **Building a benchmark?**
Run the three screens before you release it.
- ➔ **Your benchmark saturating?**
Pool the answers. No new labels.
- ➔ **Reporting abstention?**
State the response format. It moves the number by 2×.

Resources:

